HW2 - Dániel Bacsó (N6M007)

For this assignment, following the example of Hidalgo's 2007 article, I wanted to visualise the 2024 product space using OEC data. I used the HS4 level, which enabled me to visualise 1,217 product groups. To construct and filter the global product space network, I combined the Maximum Spanning Tree framework with a proximity threshold filter (0.55), which ensures a coherent structure whilst filtering out noisy, weak connections. In this network representation, the nodes correspond to products defined at the HS4 level; their size is proportional to global trade volume, whilst their colours denote the official industry sections to illustrate structural economic classifications.

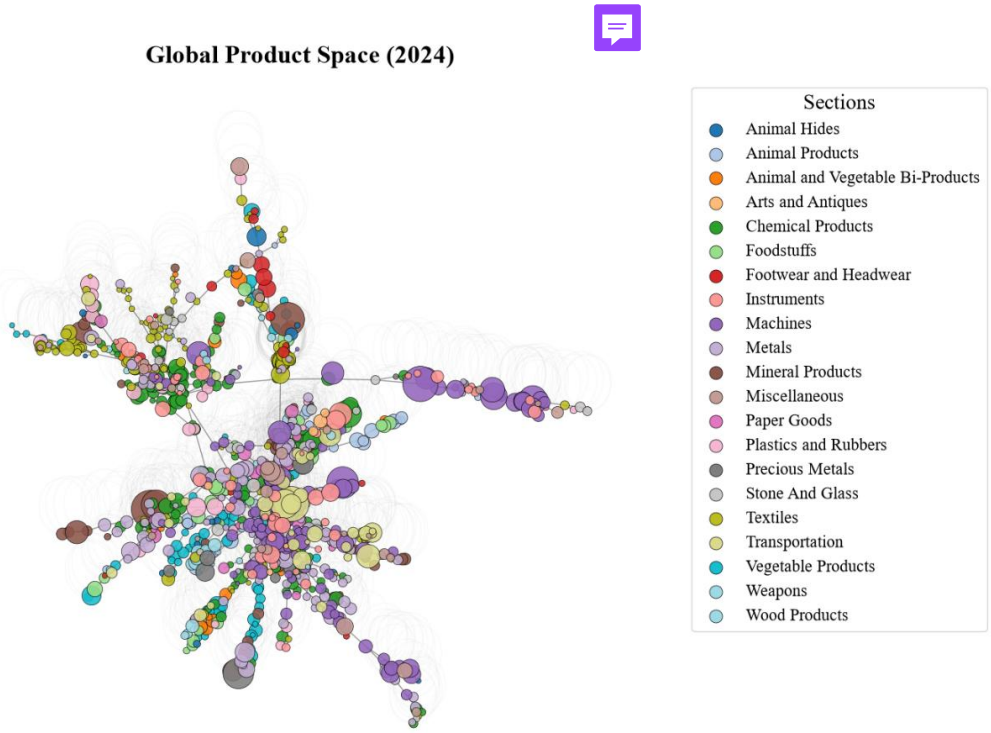


Figure 1: Global Product Space Network for 2024

Figure 1 provides a clear illustration of the core-periphery structure of international trade. The densely interconnected, higher value-added products (such as the purple and green nodes representing the engineering and chemical industries) are concentrated at the centre of the network, whilst the less complex, raw material-based goods are located at the periphery. Furthermore, the structural groupings indicate that products requiring similar technological capabilities branch out closely alongside one another within the network, forming coherent clusters.

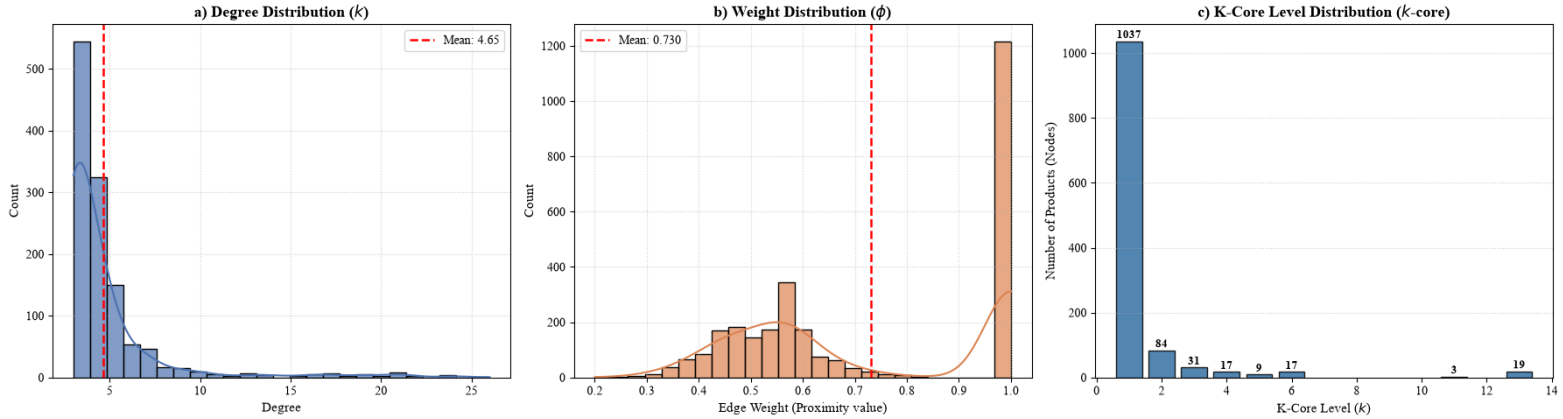


Figure 2: Structural and Statistical Distributions of the Product Space Network (2024): a) Degree Distribution (k); b) Weight Distribution (ϕ); c) K-Core Level Distribution (k-core)

Part (a) of Figure 2 clearly shows that most products have a low degree of less than 5, whilst the average degree is 4.65. However, there are also a few products (hubs) that have a degree of over 20. Part (b) of Figure 2 shows the distribution of edge weights, the mean value of which is 0.730. The distribution exhibits a notable concentration near the upper limit of 1.0. A broader range of moderate proximity values indicate that most linked product pairs exhibit capability overlap, whilst there is also a distinct layer of extremely robust technological links. Part (c) of Figure 2 illustrates the core-periphery structure of the network. Most products (1,037) are at k-core level 1, which means they form the periphery. In contrast, at the higher levels (e.g. the 19 products at level 13), a very densely interconnected, robust core emerges, where the most complex, globally central technological products are found.

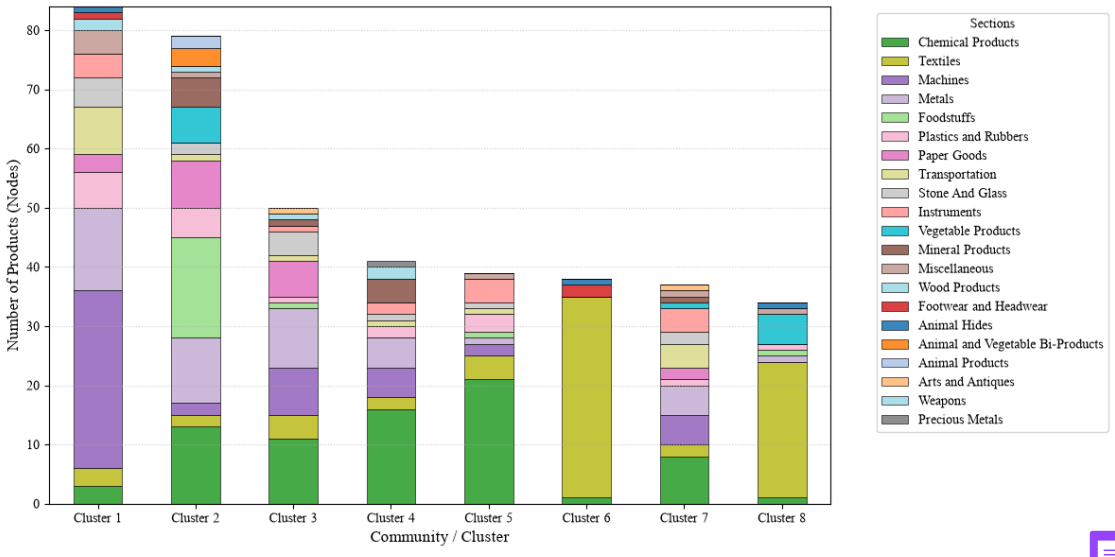


Figure 3: Industry Composition of Top Detected Communities (Louvain, 2024)

Figure 3 shows the size and sectoral composition of the eight largest communities detected by the Louvain algorithm. The data show that some clusters are highly specialised. Clusters 6 and 8 focus almost exclusively on the textile industry, whilst cluster 1 is dominated by the

engineering and metals sectors. Other communities are considerably more heterogeneous and encompass products from several different sectors.

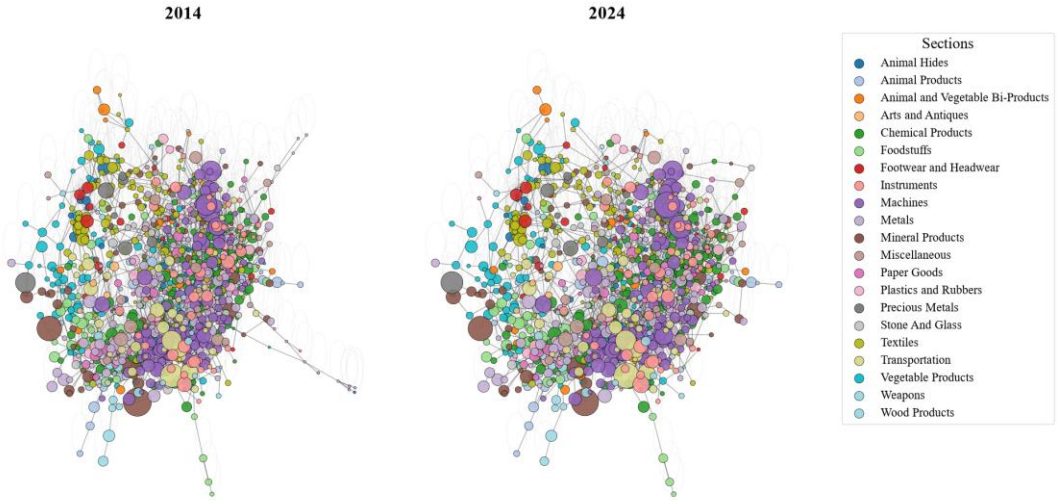


Figure 4: Dual-Year Comparison of the Global Product Space Networks (2014 vs. 2024)

A visual comparison in Figure 4 shows that the distribution of the main industrial sectors, the core-periphery arrangement and the structure of the denser technology hubs have changed very little, suggesting that the global patterns of technological capabilities and product linkages underpinning international trade will persist in the long term.

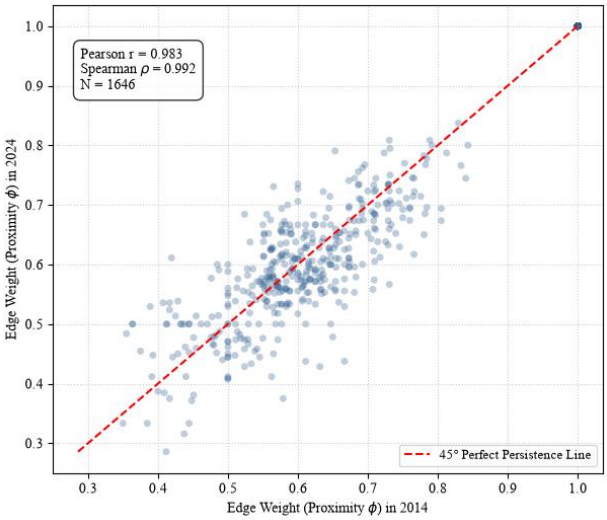


Figure 5: Persistence of Edge Weights Between 2014 and 2024

Figure 5 confirms the findings shown in Figure 4. The extremely high Pearson correlation (0.983) and Spearman's rank correlation (0.992) between the edge weights, as well as the close alignment along the 45-degree reference line, demonstrate that the technological links and overlaps in capabilities between the products have remained extremely stable over the course of the decade.